data before this crucial authentication step has completed.

In AES-SIV the MAC is AES-CMAC, which is an improved version of the AES-CBC-MAC used in COSE. Encryption is performed using AES in CTR mode. This means that AES-SIV has the same nice property as AES-CCM: it requires only an AES encryption circuit for all operations (even decryption), so can be compactly implemented.

Side-channel and fault attacks

Although SIV mode protects against accidental or deliberate misuse of nonces, it doesn't protect against all possible attacks in an IoT environment. When an attacker may have direct physical access to devices, especially where there is limited physical protection or surveillance, you may also need to consider other attacks. A *secure element* chip can provide some protection against tampering and attempts to read keys directly from memory, but keys and other secrets may also leak though many *side channels*. A side channel occurs when information about a secret can be deduced by measuring physical aspects of computations using that secret, such as the following:

- The timing of operations may reveal information about the key. Modern cryptographic implementations are designed to be *constant time* to avoid leaking information about the key in this way. Many software implementations of AES are not constant time, so alternative ciphers like ChaCha20 are often preferred for this reason.
- The amount of power used by a device may vary depending on the value of secret data it is processing. *Differential power analysis* can be used to recover secret data by examining how much power is used when processing different inputs.
- Emissions produced during processing, including electromagnetic radiation, heat, or even sounds have all been used to recover secret data from cryptographic computations.

As well as passively observing physical aspects of a device, an attacker may also directly interfere with a device in an attempt to recover secrets. In a *fault attack*, an attacker disrupts the normal functioning of a device in the hope that the faulty operation will reveal some information about secrets it is processing. For example, tweaking the power supply (known as a glitch) at a well-chosen moment might cause an algorithm to reuse a nonce, leaking information about messages or a private key. In some cases, deterministic algorithms like AES-SIV can actually make fault attacks easier for an attacker.

Protecting against side-channel and fault attacks is well beyond the scope of this book. Cryptographic libraries and devices will document if they have been designed to resist these attacks. Products may be certified against standards such as FIPS 140-2 or Commons